Question no.1:
Let f and g be the functions defined by $f(x)=3x+3 \& g(x)=3x+5$ then find $f \circ g(x)$ and $g \circ f(x)$.

Solution:

$f \circ g(x) = f(g(x))$

$= f(3x+5)$

$= 3(3x+5)+3$

$= 9x+18$

$g \circ f(x) = g(f(x))$

$= g(3x+3)$

$= 3(3x+3)+5$

$= 9x+14$

Question no.2:
Determine whether the equation has one, two or zero real solution.
$5x^2+4x-6=0$

$solution:$

$a=5, b=4, c=-6$

$b^2-4ac=16+120=136>0$

So there will be two distinct real solutions.

Question no.3:

Determine whether the equation has one, two or zero real solution.
$x^2+3x+6=0$

$solution:$

$a=1, b=3, c=6$

$b^2-4ac=9-24=-15<0$

So there will be no real solutions.

Question no.4:

The equation $tx^2+5x-6=0$ has two real solutions. What can be deduced about the value of t?

Solution:

$$tx^2+5x-6=0$$
$$a=t, b=5, c=-6$$

For two real solutions the discriminant should be >0 i.e

$$b^2-4ac>0$$
$$25+24t>0$$
$$24t>-25$$
$$t>\frac{-25}{24}$$

Question no5:

The equation $kx^2-8x+9=0$ has exactly one solution. What can be deduced about the value of k?

Solution:

$$kx^2-8x+9=0$$
$$a=k, b=-8, c=9$$

For exactly one solution the discriminant should be equal to 0 i.e

$$b^2-4ac=0$$
$$64-36k=0$$
$$64=36k$$
$$\frac{64}{36}=k$$
$$\frac{32}{18}=k$$